

Recycling of Lithium-Ion Batteries

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1 INTRODUCTION

The ongoing transition from combustion engines to electric vehicles is not only a path to reduced oil consumption but also to cleaner air in emerging mega-cities. Battery powered support in hybrid vehicles will improve driving characteristics and increase efficiency enabling techniques such as recuperation and electrical driving in slow city or traffic jam situations, where combustion engines are ineffective. In addition, the demand for lithium ion batteries (LIBs) is expected to rise in the fields of portable electronics as well as peripheral energy storage for renewable energy sources. In all fields, the electrochemical storage technology—predominantly lithium ion batteries—determines the lifetime of some products or components. These factors will even further promote a growing demand in recycling technologies of electrochemical storage devices.

Recently, Delucchi *et al.* (2014) stated that “the demand of cobalt for the production of 20 million batteries per year would be about equal to current world mine production of cobalt, and would deplete current cobalt reserves in less than 60 years (and deplete cobalt resources in about 120 years)” and that “the demand for nickel to make 20 million EV batteries per year would be two orders of magnitude larger than current world mine production.” From his data, one can deduce a 170-fold larger demand than today’s production capacities can supply (Delucchi *et al.*, 2014).

Sullivan and Simon attribute significant reductions in energy consumption for lithium-ion battery (LIB) production to the use of recycled materials^{1,2} (Sullivan and Gaines,

2012; Simon and Weil, 2013). Furthermore, the recycling of products containing Ni and Co benefits the “(1) conservation of relatively scarce or expensive materials, and (2) reduction of material production emissions like SO_x when these materials are won from earth” (Sullivan and Gaines, 2012).

Currently, recycling processes still focus on the recovery of cobalt and nickel but not lithium. This is related to the higher raw material costs compared to the less expensive lithium (Ziemann, Weil, and Schebek, 2012; Dunn *et al.*, 2012; Hanisch, Haselrieder, and Kwade, 2011a; Hoyer, Kieckhäfer, and Spengler, 2013). However, there are good reasons for developing recycling processes that are capable of regaining lithium as well. Weil *et al.* find that no real lithium scarcity is foreseen until 2050 when easily extractable reserves in stable countries could decrease significantly (Weil and Ziemann, 2014). Kushnir and Sandén (2012) highlight that recent debates often conclude there is sufficient lithium available in terms of resources, but that the “flowrate into society” may be more critical. The lithium resources excluding the ocean, calculated at 30 Mt, by Kushnir *et al.* “will only be exhausted this century if large scale use of predominantly EV sized batteries comes into play, or if batteries are not recycled. For all EV scenarios or PHEV scenarios not involving a drastic reduction in mobility trends, recycling will be a necessary option if not for the build-up, then for maintenance of a societal lithium stock. Yet recycling economics are currently not good and may actually degrade somewhat instead of improving” (Kushnir and Sandén, 2012). The possibility of “degrading” is associated with an estimated substitution of valuable metals such as Co and Ni in LIB materials and ultimately to the loss of economic attractiveness for recyclers (Kushnir and Sandén, 2012; Wang *et al.*, 2014). Gruber *et al.* (2011)

estimate the resources of lithium at 38.68 Mt but agree that lithium has to be recycled with recycling rates of at least 90% to prevent a future shortage (Gruber *et al.*, 2011). Miedema and Moll (2013) postulate that recycling is the most important source of lithium for the European Union on a long-term basis. In optimistic scenarios, short-term lithium availability could even limit the growth of electric mobility (Miedema and Moll, 2013). Furthermore, Zeng and Li (2013) see a necessity of a lithium recycling rate of at least 90% in China (Zeng and Li, 2013) and Wang *et al.* (2014) estimate the lithium recycling efficiency at 80% for the United States, when commenting on economies of scale for future LIB recycling infrastructure (Wang *et al.*, 2014). To encourage investments in recycling processes and to make them economically feasible sooner, Hoyer, Kieckhäfer, and Spengler (2013) recommend that political decision makers should allow “temporal storage of intermediates, e.g. cells and cathode coating, and alter the associated calculation method of the recycling rate in a way that recycling deficits from early years could be compensated in later years when substantial battery return flows are reached” (Hoyer, Kieckhäfer, and Spengler, 2013).

A broad majority of experts deduce from aforementioned facts that a future shortage of lithium and other valuable raw materials used in LIBs can only be prevented by further development and broad application of the state-of-the-art recycling processes including lithium recycling. Thus, lithium-rich active materials recovered from LIBs need to be of sufficiently high quality and purity levels. In one possibility, the developed recycling processes can be designed to regain highly concentrated suspensions or powder fractions of active materials, which can be used as reactants in metallurgical processes to recover lithium salts (LiOH , Li_2CO_3) and transition metal (Co, Ni) products (Hanisch, Haselrieder, and Kwade, 2011a; Krüger *et al.*, 2014). Alternately, electrode production rejects and eventually also cycled active materials should be recovered for a direct reuse in new LIBs for automotive applications (Krüger *et al.*, 2014).

A good overview about the recycling of LIBs is presented by Vezzini (2014), Bernardes, Espinosa, and Tenório (2004), Espinosa, Bernardes, and Tenório (2004), Zeng, Li, and Singh (2014), Zhang *et al.* (2013a), Xu *et al.* (2008), Wen *et al.* (2012a), Swain *et al.* (2007), Granata *et al.* (2012a), Georgi-Maschler *et al.* (2012), and Vezzini (2014).

2 LITHIUM-ION BATTERIES

Secondary LIBs consist of several LIB cells controlled by a battery management unit (BMU) to prevent deep discharge or overcharging and to monitor the cells' state of health.

Bigger battery systems, for example, for electric vehicles (EVs), have another subsystem level between system and cell, the modules or stacks. From the recycler's point of view, these battery systems (without the battery cells) can contribute to a recycling rate of up to 50% mass fraction to fulfill legislative requirements and to pay off the recycling process. This is because peripheral components such as cables, busbars, casings, and electric components contain a combination of copper, steel, aluminum, and plastic in addition to the battery cells. All of these peripheral components can be recovered easily using conventional waste electrical and electronic equipment (WEEE) recycling methods.

The battery cell itself consists of several galvanic elements between the cathodes and the anodes, which are wound or stacked and electrically isolated by an ion-permeable separator. The cathode is composed of an aluminum foil (current collector) coated with an electrochemically active material. Often, a lithium-transition-metal-oxide LiMO_2 (M stands for metals such as Co, Ni, Mn, Al, or their combinations, e.g., $\text{LiNi}_{0.33}\text{Mn}_{0.33}\text{Co}_{0.33}\text{O}_2$) are used in automotive applications. Driven by the prices of these transition metals, material manufacturers are working on the reduction and finally on the substitution of cobalt and nickel in battery active materials (see Figure 1). The current prices of LIBs are still too high for a broader use in electric mobility. The next generation of 5V-lithium-transition-metal-oxides will have lower Co content yet still a significant Ni content, for instance. Further spinel or olivine structures, with low Ni content and even without Co such as LiMnNiO_4 and LiFePO_4 , have high potentials to replace materials such as $\text{LiNi}_{0.33}\text{Mn}_{0.33}\text{Co}_{0.33}\text{O}_2$ in the future. Further future lithium-based battery generations, such as lithium sulfur (LiS) or lithium metal batteries, will not require any transition metals. This is obviously a great challenge for the design of recycling processes, presently relying on the economic contribution of the more valuable elements Co and Ni. The anode is a copper foil coated with graphite.

The adhesion toward the current collector and inside the particulate coating of these electrodes is facilitated by a polymeric binder, mostly polyvinylidene fluoride (PVDF). On the anode side, carboxymethyl cellulose (CMC) and styrene-butadiene rubber (SBR) are also used as binders. In addition, some carbon black improves the electric conductivity in both anodes and cathodes.

The ion conductivity is facilitated by an electrolyte. Usually, a mixture of two or more organic solvents based on ethylene carbonate (EC), dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), and others with a solute lithium-rich conductivity salt, such as LiPF_6 , is employed. The cell casing is usually made of aluminum or steel in the case of cylindrical or prismatic wound cells. For stacked cell types, a soft case made of aluminum-polymer compound foil, the so-called coffee bag, is used as cell casing.

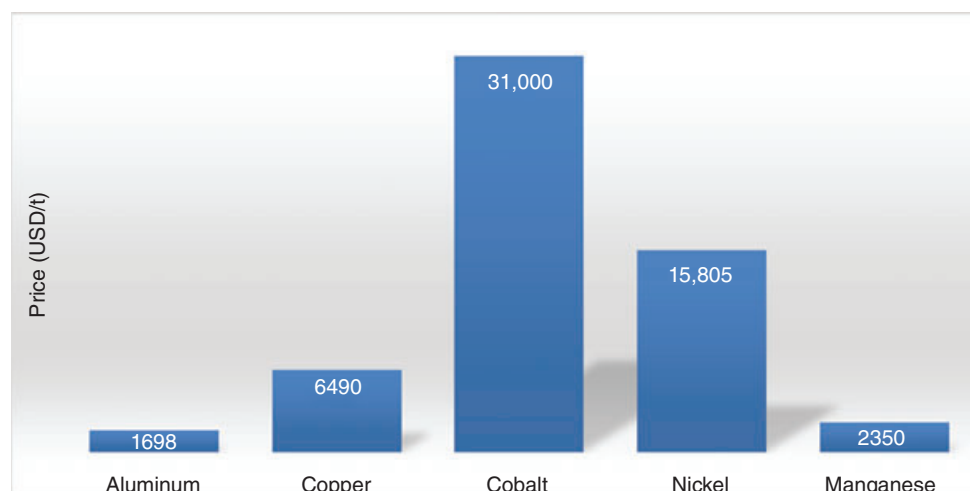


Figure 1. Prices of metals from NMC-battery electrodes (date: 14 March 2014).

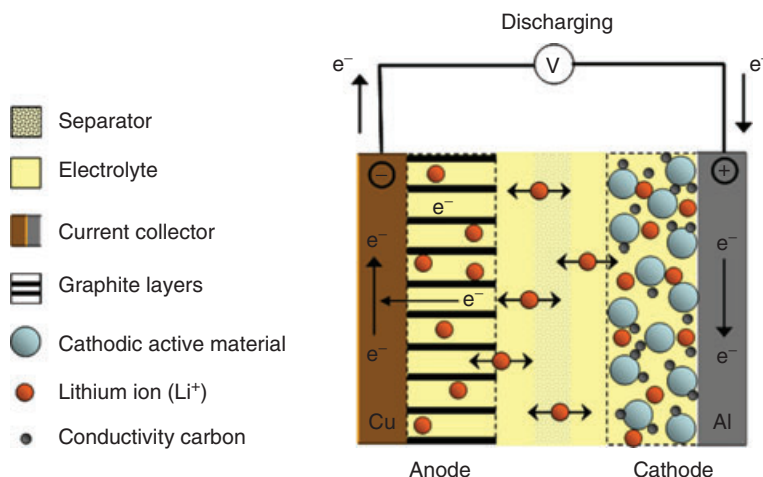


Figure 2. Schematic function of a lithium-ion battery. Source: Reproduced from Hanisch *et al.*, 2014. © TU Braunschweig.

When a battery is discharged, electrons move from the anode to the cathode via the external electrical circuit, and electrical work is executed at the load (see Figure 2). More precisely, the electrons participate in electrode reactions at the solid–electrolyte interface inside the electrolyte-wetted electrode coating. Therefore, the electrode composite structure has to be electrically conductive, which is enabled by conductivity additives such as carbon blacks (Daniel and Besenhard, 2011; Bockholt, Haselrieder, and Kwade, 2013), and a defined compression of the electrodes should be applied (calendering) (Haselrieder *et al.*, 2013, 2014; Tran *et al.*, 2012). Further information can be found in the publications of Yoshio, Brodd, and Kozawa (2009) and Wakihara and Yamamoto (1998). The induced potential difference is the driving force for the movement of the lithium ions toward

the positively charged cathode. Lithium ions are inserted into the lattice host of the cathode materials, for example, LiMO_2 . During the discharge process, active materials of the anode and cathode are oxidized and reduced, respectively. Ion conduction between and inside the particulate electrode pore structure as well as inside the separator is carried out by the electrolyte. Furthermore, anodic and cathodic compartments are electrically uncoupled by the separator. During the charge process, ion and electron transport is going on in the opposite direction. The principle of a secondary lithium ion battery is based on the reversible insertion and extraction of lithium ions into the active materials of cathode and anode.

Aging effects result from electrochemical, chemical, and mechanical processes during charging and discharging

(Vetter *et al.*, 2005; Vanimisetti and Ramakrishnan, 2011; Zhang *et al.*, 2000; Kanevskii and Dubasova, 2005). These effects influence the capacity and performance of secondary LIBs. Owing to the requirement of a life of 10 to 15 years or 4000 to 5000 cycles (at least in EVs), these effects are of high consequence and have to be minimized. And furthermore, sooner or later, these batteries will have to be recycled.

3 UNIT OPERATIONS IN BATTERY RECYCLING

Recycling processes for LIBs are combinations of different unit operations. At first, a deactivation step should be chosen to lower the dangers resulting from the stored energy and chemical reaction potential of charged LIBs. These can be thermal pretreatment steps to volatilize the electrolyte or even to decompose all organic compounds of the battery cells. In one method, a discharging step can be used to reduce the hazard level due to the reduction of stored energy, which could otherwise activate further reactions. Alternatively, the electrolyte can be frozen to prevent electrochemical reactions and short circuits during a subsequent crushing process.

The following unit operations can be divided into mechanical, pyrometallurgical, and hydrometallurgical treatments. Mechanical treatment implies crushing to open battery cells or modules followed by the liberation of the valuable materials, which is accompanied by classifying and sorting processes to separate copper foil, aluminum foil, separator, and the coating materials. In addition, the casing of the battery cells, modules, and connecting components can be recovered. During pyrometallurgical processes, several components of battery cells or the battery cells themselves are melted. While the transition metals nickel, cobalt, and copper can be recycled from the cast, lithium and aluminum remain in the slag. Further treatment is necessary for the recovery of lithium. This can be a hydrometallurgical process that includes leaching of the educts, extraction of the valuable products, crystallization, and precipitation. Hydrometallurgical treatment is used to recover pure metals, for example, lithium, from separated coating materials derived from mechanical processes or from slag derived from pyrometallurgical processes.

Figure 3 summarizes possible process options for the involved unit operations such as deactivation, mechanical treatment, as well as the subsequent pyro- and hydrometallurgical treatments. The scheme displayed in Figure 3 depicts the advantage of combining different unit operations in order to fulfill the goal of high recovery rates. Thus, the casing, copper and aluminum foil, and the most

valuable components such as nickel, cobalt, and manganese can be recovered using a combination of mechanical and hydrometallurgical treatments. Further information about unit operations for recycling of LIBs is summarized by Vezzini (2014), Bernardes, Espinosa, and Tenório (2004), Espinosa, Bernardes, and Tenório (2004), Zeng, Li, and Singh (2014), Zhang *et al.* (2013a), Xu *et al.* (2008), Wen *et al.* (2012a), Swain *et al.* (2007), Granata *et al.* (2012a), and Georgi-Maschler *et al.* (2012).

3.1 Deactivation

For “end-of-life” transport of damaged LIBs or as a pretreatment step before recycling, deactivation is necessary. During such a treatment, the electrochemically stored energy is reduced and, therefore, the possibility of a thermal runaway and other chemical reactions are minimized. In the recycling of LIBs, the application of several different treatments is possible.

3.1.1 Discharge

The discharge and charge operations during the life span of a LIB in an EV or hybrid electric vehicle (HEV) are normally controlled by a BMU. The main functions of this unit are monitoring the voltage of each battery cell or module, controlling the current flow during charging and discharging, detecting temperatures, and taking suitable countermeasures in case of malfunctions (Korthauer, 2013). In addition, the BMU controls the end-of-charge and the end-of-discharge voltage. Charging and discharging by the BMU is only possible in a specific voltage range specified for the individual battery chemistry. For a secure disassembly of battery systems or battery modules and the following process steps, a complete discharge of the energy stored is preferable. To accomplish this, voltages beyond the default end-of-discharge voltage are needed. This adjustment is often prevented by the BMU. Such a deep discharge could be accompanied by a polarity reversal and formation of flammable gases in the cell (Mähliß, 2012). The avoidance of rising inner pressure within a battery cell due to gas formation can be achieved by specific discharge techniques. The literature describes the phenomenon of voltage relaxation of LIBs (Bernardi and Go, 2011; Fuller, Doyle, and Newman, 1994; Roscher, Bohlen, and Vetter, 2011). The probability of a voltage increase after a deep discharge is very high. Thus, direct short-circuiting of the battery system, module, or cell is advisable. Discharging followed by short-circuiting finally decreases the probability of exothermal reactions of a damaged or overheated battery cell (Section 4).

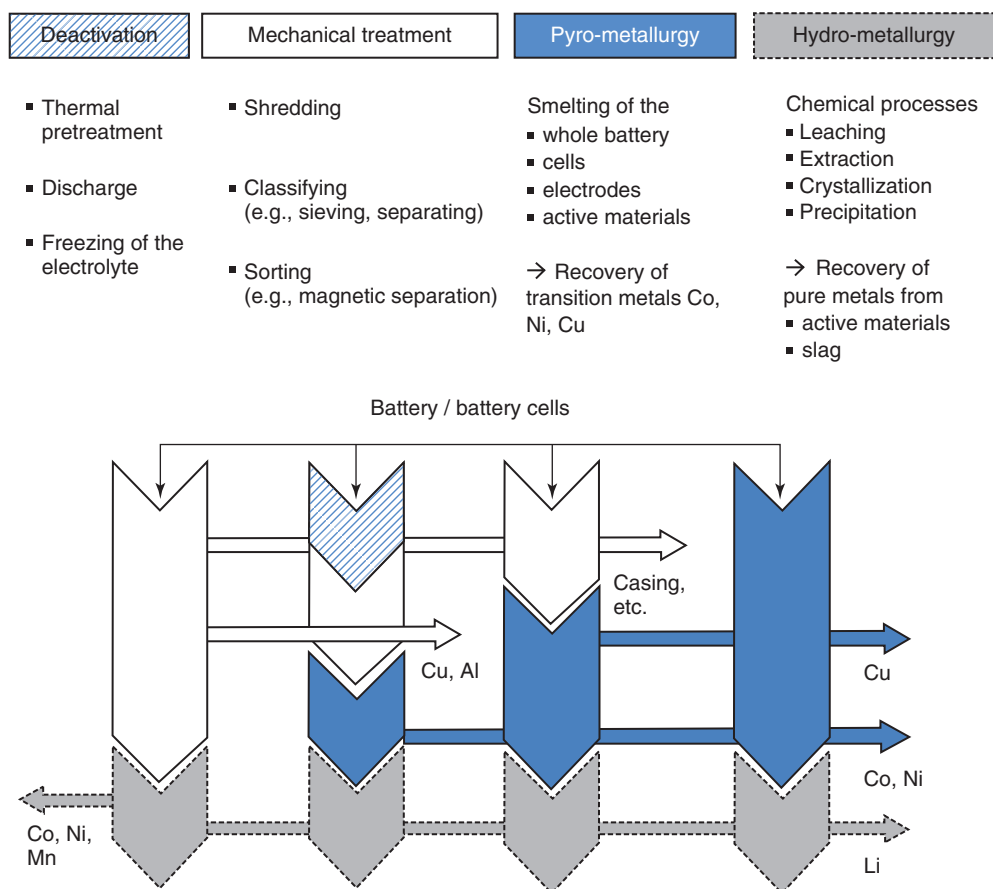


Figure 3. Unit operations in battery recycling and their possible combinations to establish efficient recycling process routes.

3.1.2 Freezing

Another method to decrease the potential of exothermal reactions due to rising temperature is freezing damaged or defective LIB cells. Here, a freezing of battery systems, modules, or cells down to -65°C is recommended (BAM Bundesanstalt für Materialforschung und -prüfung, 2013). The low temperature fixes hazardous materials, and chemical reactions are obviated. This treatment can be used as a pretreatment for transportation of damaged batteries. During transport, a maximum temperature of -60°C of the battery cells has to be ensured and interfaces should be secured (BAM Bundesanstalt für Materialforschung und -prüfung, 2013). Freezing is also possible as a pretreatment for mechanical recycling processes such as crushing. It suppresses the heat generation by mechanical strain, lowers the reaction potential, and simultaneously achieves an embrittlement for a more efficient crushing and separation process. Enhanced safety can only be accomplished by freezing if the materials are continuously kept at low temperatures. Therefore, a cold chain or a

combination with another deactivation method should be provided.

3.1.3 Thermal treatment

The Belgian company Umicore uses a thermal deactivation in a furnace as the first step in their Val'Eas® recycling process (Section 5). In a preheating section, the battery cells are slowly heated up to maximum temperatures of 300°C . At these temperatures, chemical reactions occur inside the cell and the solvents of the electrolyte evaporate. Owing to the rise in the inner pressure, the cells vent and the gaseous compounds exhaust and are inflamed. After this treatment, the battery cells are electrochemically inactivated and free of flammable solvents (Vezzini, 2014). Because of the gaseous combustion products, exhaust after-treatment is needed.

Accurec, a German battery recycler, utilizes a thermal deactivation process in a vacuum chamber with maximum temperatures of 250°C and a vacuum of approximately 700 mbar. The electrolyte evaporates and condenses in a downstream condenser. This treatment deactivates the

battery cells and recovers the electrolyte, but this product still contains impurities caused by decomposition products (Georgi-Maschler, 2009).

3.1.4 Discharge in conductive liquid

For transport, or as a pretreatment step, LIBs could be discharged by electrolysis of a conductive liquid medium. For this treatment, an aqueous sodium chloride solution with a concentration of 0.3% by weight is recommended (BAM Bundesanstalt für Materialforschung und -prüfung, 2013). The battery system is placed in this solution inside a nonconductive drum. A second container should be placed around it to prevent leakage. The discharge and the storage of the battery system have to be controlled for at least 24 h with respect to the voltage and the temperature. During the electrolysis, gas formation occurs as shown in Figure 4. In addition to reactions of ions within the solution, reactions can also take place between these ions and the battery casing. Consequently, this could lead to an opening of the cell casing due to corrosion followed by a release of electrolyte and a reaction between the conducting salt and the saline solution. This will then result in the formation of reactive and acidic compounds such as hydrogen fluoride. Using sodium hydroxide dissolution instead of sodium chloride, the acidic compounds are neutralized.

3.2 Disassembly of battery systems

EVs, plug-in hybrid electric vehicles (PHEVs), and HEVs contain LIB cells, assembled in battery modules, which are the subunits of the battery systems as depicted in Figure 5. Additional components are the BMUs, the sub-BMUs on module level (Herrmann *et al.*, 2012), and further electronics (e.g., temperature sensors as well as the casing

and connecting elements). A disassembly process of the battery systems is one of the first steps in the recycling process. The process step harbors different problematic tasks because electric vehicle battery systems have a high weight combined with high voltage and complex structure (Herrmann *et al.*, 2012). The electronic elements (e.g., the BMU) and the system casing can be separated and recycled in existing processes for consumer electronics. This disassembly process decreases the size and the electrochemically stored energy content of the starting material, for example, for crushing or melting processes, thus reducing safety issues and impurities in the following recycling process. Recovering all components, including the materials of the battery cells, is the goal.

At present, the variety of battery system designs is very high because of a lack of standardization. Consequently, a high input of manual disassembly steps is needed (Herrmann *et al.*, 2014a). Owing to the high voltage, only electricians are allowed to handle charged systems, which significantly impacts labor costs. Therefore, a discharge of the whole battery system before disassembly is beneficial. In the future, as the standardization of battery system structure and the total number of systems to be recycled increase, in this scenario, automation of the disassembly will decrease the costs, mainly labor costs, of the process. Schmitt *et al.* developed a flexible gripper to remove battery cells and simultaneously characterize battery cells voltage and impedance (Schmitt *et al.*, 2011a). Furthermore, Herrmann introduced a disassembly process combining manual and automated process steps (Herrmann *et al.*, 2012). This includes a manual dismantling after discharging of the battery system and automated removal of battery modules and battery cells subsequently. In addition, hybrid working stations, where technicians and robots work next to each other, could be a potential approach to increase disassembly efficiency. In this case,

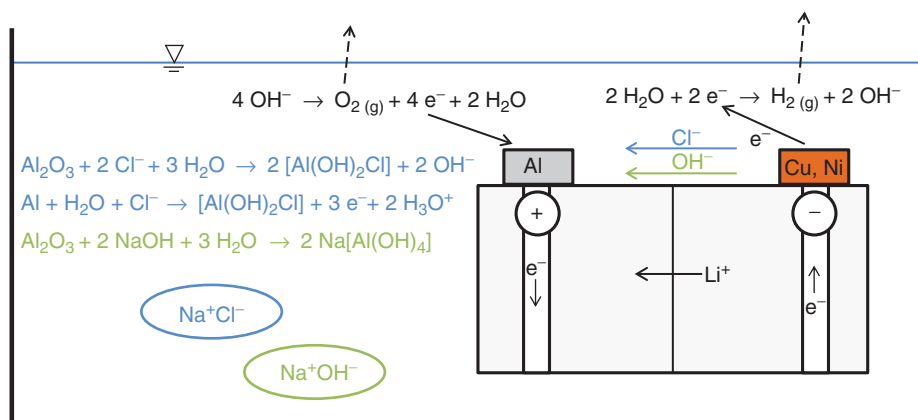


Figure 4. Reactions and gas formation during electrolysis of a sodium chloride and a sodium hydroxide dissolution.

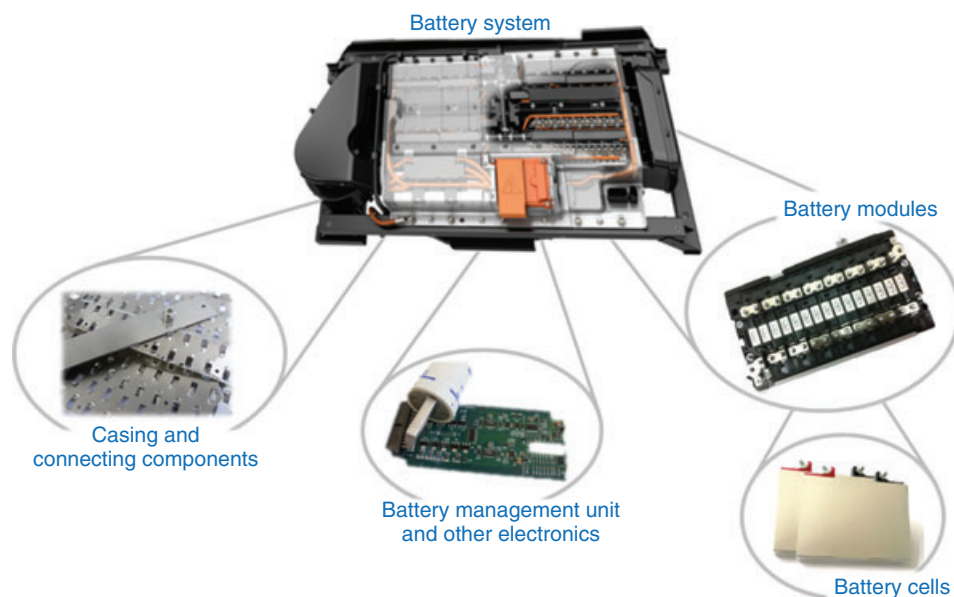


Figure 5. Components of a battery system.

the robots can be programmed to fulfill specific disassembly steps that are specific to the battery systems due to the absence of standardized battery system designs. This is an approach to decrease the overall costs of the disassembly processes (Herrmann *et al.*, 2014a).

3.3 Crushing

After the disassembly of LIB systems into smaller subunits, the next step of a recycling process is crushing in most cases. Within this process step, the opening of the battery cells takes place and valuable components of the cells are released. In addition, the battery packs are converted into a product flow that could be processed, transported, and stored as bulk material. Depending on the starting material, the following specific crushing methods can be distinguished.

3.3.1 Wet crushing

The company Retrie Technologies Inc. realized an automated shredding process especially designed for LIBs. For crushing, a hammer mill is operated in a brine solution (Vezzini, 2014). Usage of this dissolution neutralizes acid and other emissions of chemical reactions, which occur during crushing. After this mechanical treatment, solids and liquids can be separated.

3.3.2 Inert crushing

Zhang *et al.* studied the characteristics of dry and wet crushing of spent LIBs. They concluded that dry crushing

shows advantages because of fewer impurities in the fine fraction. Furthermore, a better separation of metal electrode substrates and the particulate coating components can be achieved (Zhang *et al.*, 2013b). When it comes to dry crushing, several characteristics of LIBs have to be taken into account. The electrolyte (10–20 wt% of EV or HEV battery cells) contains flammable organic solvents such as ethylene carbonate (EC), dimethyl carbonate (DMC), and ethyl methyl carbonate (EMC). These solvents form an explosive mixture with aerial oxygen. Other components such as the commonly used binder polyvinylidene fluoride (PVDF) and lithium-based conducting salts containing fluorine decompose at elevated temperatures, thereby forming acid gas compounds such as hydrogen fluoride and other gaseous products (Section 4). An ignition of the atmosphere inside a shredder has to be avoided. The company Hosokawa Alpine AG developed a gastight shredder for dry inert crushing of LIB modules in the funded German research project LithoRec II. This shredder consists of a single rotary shear with forced feeding. It is equipped with a discharge grate that allows upper particle size limitation and thereby an adjustable degree of liberation. For establishing inert conditions inside the shredder, the oxygen content is reduced to avoid formation of explosive mixtures of organic carbonates from the electrolyte and aerial oxygen. The product discharge is performed by a gastight connection to the following process step. To avoid exothermal chemical reactions, dry crushing processes should be fed with completely discharged starting material (battery cells or modules).

A thermal pretreatment can be used, for instance, to deactivate the battery cells and to remove reactive and flammable electrolyte in advance (Georgi-Maschler, 2009). Therefore, no actions concerning these matters are necessary. However, the dry carbonaceous coating has the potential to provoke dust explosions depending on the process design and the starting material. Dust emissions must also be avoided.

3.4 Sorting, sieving, and compound separation processes

After crushing the resulting fragments of electrodes, separator, case, and the powder fraction, consisting of coating agglomerates, have to be separated for further recycling. This can be accomplished with a combination of sorting and sieving steps.

At first, ferromagnetic parts can be separated by a magnetic separator. The further separation of the lasting heavy fraction can be accomplished by air separation methods such as cross-flow classification or zigzag sifting. In a zigzag sifter, the mixed materials are fed to the zigzag tube at a certain height. Materials are uplifted to the top of the sifter by an adjustable airstream and to the bottom

by gravity. Thus, the zigzag geometry enables a multistage cross-flow separation—with one cross-flow separation in each zig and zag—resulting in a heavy fraction leaving the bottom of the sifter and a lightweight fraction exiting at the top (Tomas and Gröger, 2001). A schematic of the zigzag air classification process and the gained fractions are shown in Figure 6. The light fraction consists of separator, smaller foil fragments, and already detached coating agglomerates, whereas the heavy fraction contains aluminum and copper foils still partly coated with graphite and active material (for details, see Figure 7). After some classification steps, a second, but more intensive, milling step can be applied to remove the remaining particulate coating from the current collector foil. This increases the yield of coating fragments and foil with less impurities in each of the fractions (Hanisch, Haselrieder, and Kwade, 2011b). Alternatively, other mechanisms such as dissolving the binder in a solvent (Hanisch, Haselrieder, and Kwade, 2011a; Contestabile, Panero, and Scrosati, 2001; Li *et al.*, 2009a, 2013) or decomposing the binder under high temperatures (Hanisch, Haselrieder, and Kwade, 2013) are possible process steps to separate the current collector and the particulate coating.

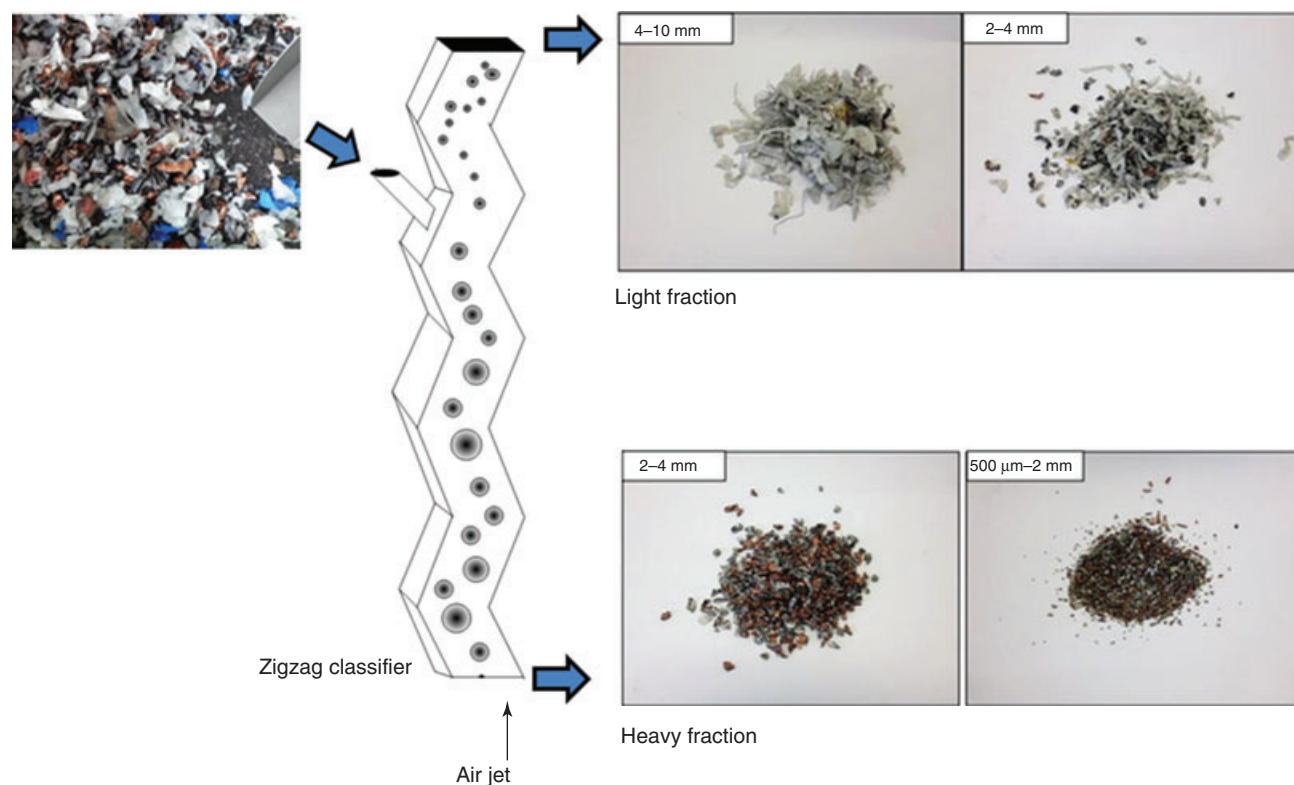


Figure 6. Air classification in recycling processes for lithium-ion batteries using a zigzag classifier; typical light- and heavyweight fractions are shown.

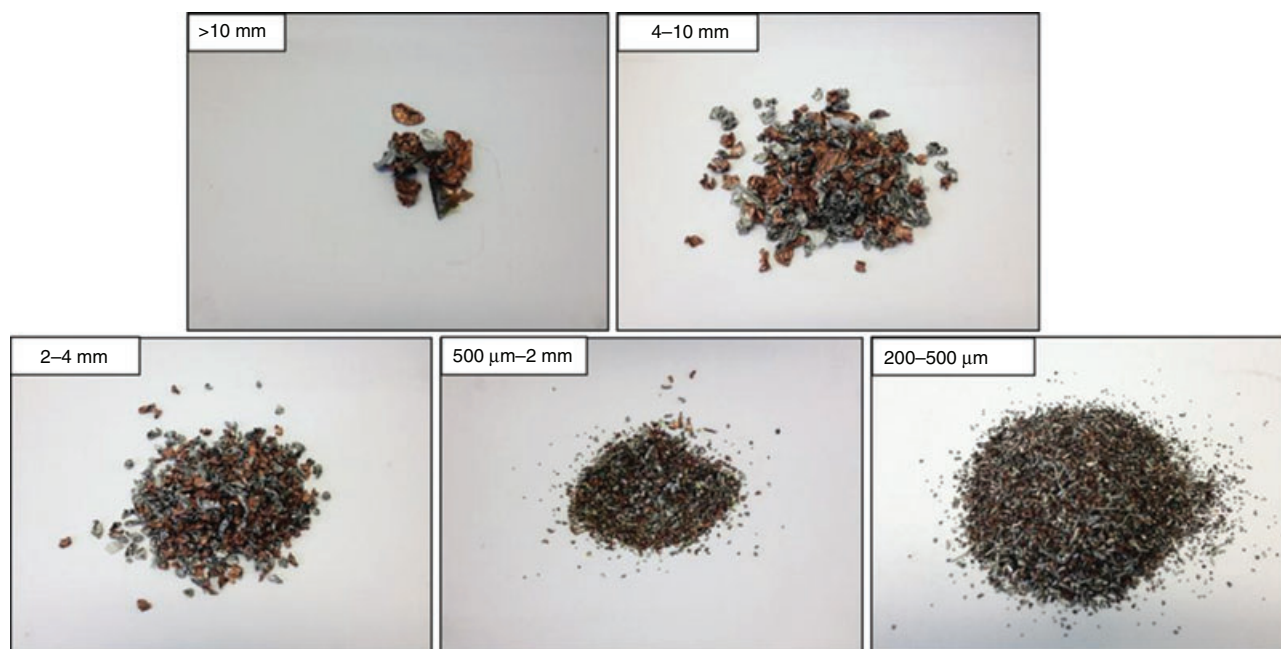


Figure 7. Sieved heavy fractions—partly coated electrode foils (Al and Cu).

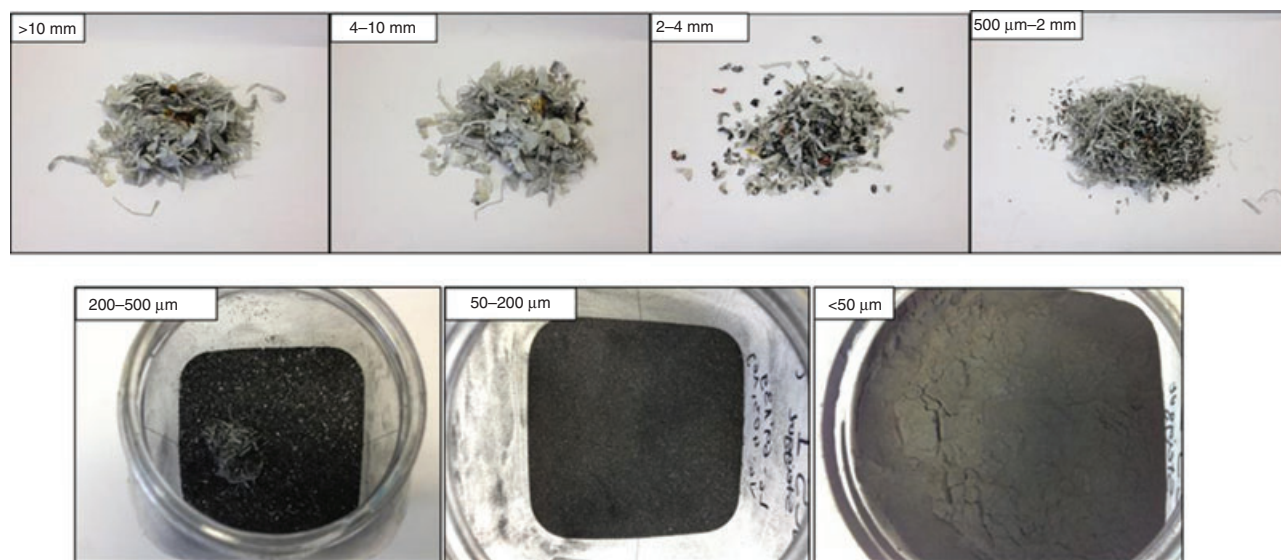


Figure 8. Sieved lightweight fractions—separator and coating agglomerates.

Additional sieving steps can be applied to regain a light- and a heavy-weight powder fraction of less impurity. Because the mechanical stressing not only loosened the coating from the foil but also resulted unwillingly in some smaller foil fragments, these fragments have to be classified, for example, by the use of a sieve with a small mesh size of approximately 50–200 μm . The finer the mesh size, the smaller the scraps that pass the sieve and the lower

the impurities of each fraction. These impurities can be quantified via atomic absorption spectroscopy (AAS) with high accuracy. Differences in the range of percentages can also be distinguished visually (see Figures 7 and 8). The regained powder fraction still contains all components of the original coating: the desired lithium-rich battery active material, graphite, conductive carbon, binder, and even some components from the electrolyte like organic residues such

as ethylene carbonate, dimethyl carbonate, and conductivity salt. These organic fractions could disturb following hydrometallurgical processes; therefore, diminishing their concentration becomes an important process task.

The separation of organic components can be accomplished partly by dry sifting processes if the components are not connected in agglomerates. If the coating still consists of agglomerates, in principal, a certain separation of anode and cathode coating agglomerates—but not of single components of each electrode compound—is possible. However, such a separation is feasible if the mechanical stressing process results in anode and cathode coating agglomerates with very similar and narrow particle size distribution. For advanced separation, the single components have to be deagglomerated by thermal decomposition (Hanisch, Haselrieder, and Kwade, 2011b), or alternatively, the organic components can be calcinated directly (Georgi-Maschler *et al.*, 2012; Sun and Qiu, 2011; Lu *et al.*, 2013). In a patented process by Retrie Technologies, the coating is calcinated, and the organic residuals are separated using a floatation process (Smith and Swoffer, 2013).

One possible process chain to combine several separation processes is shown in Figure 9. A first step can be an inert precrushing process combined with a condensation of

organic carbonates from the gas flow. The resulting scraps could optionally be stirred in an extraction reactor to extract the conductivity salt in order to regain it and to reduce the hazard level of the scraps. A following dryer reduces the content of organic carbons. Ideally, these extracted solvents are regained in a drying gas condensation step. A magnetic separation step separates the ferromagnetic fraction out of the electrode scraps, which is fed into a zigzag classifier afterward. Separator and some already separated coating agglomerates will leave the classifier in the lightweight fraction at the top while still partly coated scraps of cathodes and anodes are the heavy fraction. These scraps are heated to the decomposition temperature of the binder in a furnace and the scraps, with a lowered coating adhesion, are separated in an impact liberator (ANVIL, see below and Hanisch *et al.*, 2014a). Afterwards, pure aluminum and copper foils are regained from the sieve and the coating particles are separated from the air stream by a cyclone (Hanisch, Haselrieder, and Kwade, 2011b).

3.4.1 Special case: thermomechanical separation

Thermomechanical separation is a new technique to separate current collector foil and coating. It is a combination of a

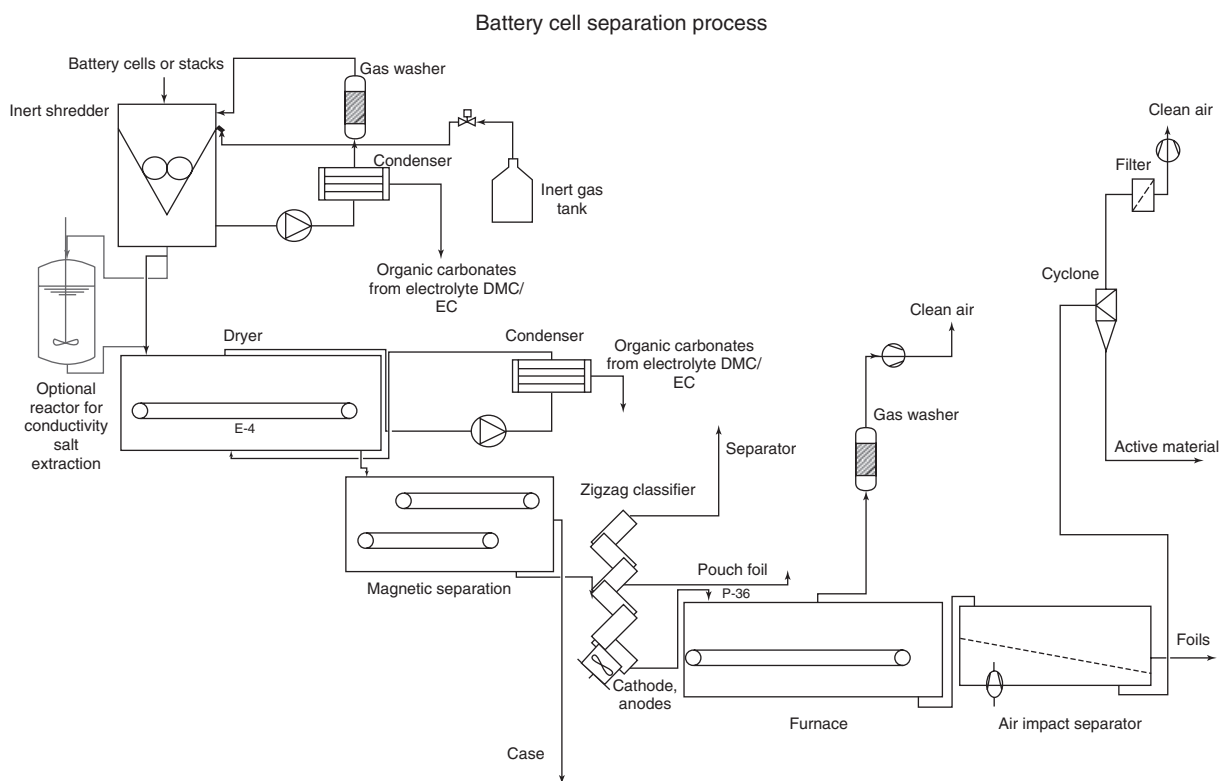


Figure 9. Process chain to separate different components from lithium-ion batteries. *Source:* Reproduced from Hanisch *et al.*, 2011b.

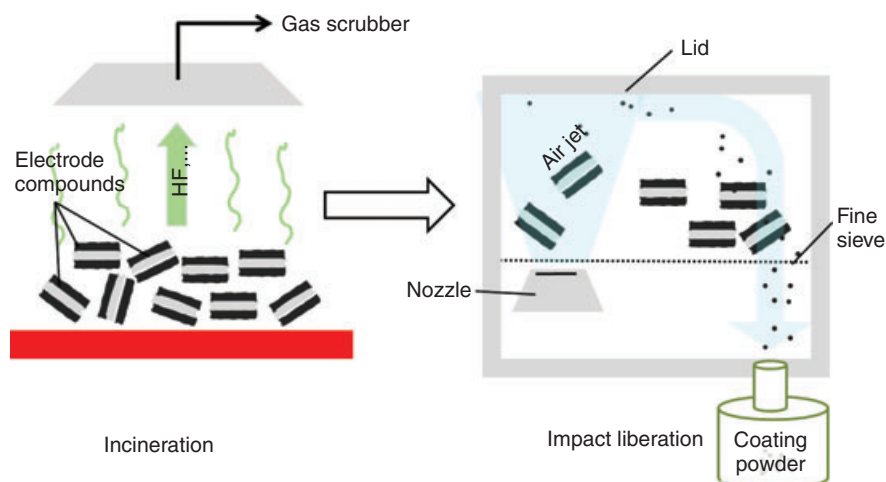


Figure 10. Schematic thermomechanical separation process for lithium-ion battery electrodes. *Source:* Reproduced from Hanisch *et al.*, 2011b.

thermal decomposition process of the binder polyvinylidene fluoride (PVDF), impact stressing, and air jet sieving.

Depending on the binder chemistry, the electrode scrap is heated up to 400–800°C and decomposes with release of hydrogen fluoride and other toxic components. These combustion gases are the drawback of this process combination. They have to be analyzed and scrubbed carefully to prevent negative impacts on workers and the environment.

Thereafter, the heat-treated scraps of electrodes are fed into the separation chamber of an air impact separator—a modified air jet sieve. An air jet accelerates the scraps to the lid of the chamber where an impact loosens the coating and breaks coating agglomerates. These smaller coating agglomerates can pass a fine sieve at the bottom of the separation chamber and are regained from the air stream by a cyclone or a filter (see Figure 10) (Hanisch, Haselrieder, and Kwade, 2011a, 2011b; Hanisch *et al.*, 2014a). This set up—developed by TU Braunschweig and Lion Engineering GmbH—results in high separation efficiencies of over 99% w/w of the coating and very low impurities from current collector foils in the regained coating fraction of less than 0.1% w/w. The purity of the regained fraction is critical because it influences the quality of potential recycling products from metallurgical steps (see Section 3.5 and Krüger *et al.*, 2014).

Sun and Qiu used vacuum pyrolysis for 30 min at 600°C with a pressure below 1 kPa to evaporate organic components (Sun and Qiu, 2011). Other processes using thermal decomposition of the binder components are discussed by Li *et al.* (2009a, 2013), Sun and Qiu (2011, 2012), Lu *et al.* (2013), Song *et al.* (2013), Lee and Rhee (2002), and Gu and Nie (2011).

3.5 Hydrometallurgical processes

In order to regain the valuable transition metals Co and Ni as well as Li from separated coatings, hydrometallurgical processes such as leaching, extraction, crystallization, and precipitation can be applied. Furthermore, products of the hydrometallurgical processes can be used to resynthesize battery active materials in battery grade. More comprehensive reviews on hydrometallurgical processes in the field of recycling of LIBs are by Chagnes and Pospiech (2013) and Joulié, Laucournet, and Billy (2014) and included in general reviews by Zhang *et al.* (2013a), Xu *et al.* (2008), and Zeng, Li, and Singh (2014).

3.5.1 Leaching

The first step of a hydrometallurgical treatment is leaching of the regained powder fractions including the active material fraction. In this process, the lithium-transition metal compounds are leached in inorganic or organic acids in order to prepare for the later separation steps. The reduction of impurities and organic residues and the separation of the different product metals, as purely as possible, are the goals of the leaching process step.

Lee uses nitric acid (HNO_3) under addition of hydrogen peroxide (H_2O_2) to leach LiCoO_2 . He studies the influence of the process parameters concentration, solid–liquid ratio (S:L), temperature, and the amount of H_2O_2 added. By addition of hydrogen peroxide to the nitric acid used for leaching, Co and Li extraction is increased from 40% to 85% and 75% to 85%, respectively. Lee explains this effect by the reduction of Co^{3+} to Co^{2+} , which can be readily dissolved. He states further an “increased extraction of Co and Li with

increasing HNO_3 concentration, temperature and the amount of hydrogen peroxide and with decreasing S:L ratio." The optimum leaching conditions found by Lee are 1 M HNO_3 , 20 g/L initial S:L ratio, 75°C, and 1.7 vol% H_2O_2 for 1 h of process time (Lee and Rhee, 2002).

Li et al. compared leaching processes using sulfuric acid (H_2SO_4), hydrochloric acid (HCl), and citric acid ($\text{C}_6\text{H}_8\text{O}_7 \cdot \text{H}_2\text{O}$). They found the processes to be affected positively by ultrasonic assistance in 2014 (Li et al., 2014). In 2013, they focused on organic acids as leaching reagents by comparing citric acid, malic acid ($\text{C}_4\text{H}_6\text{O}_5$), and aspartic acid ($\text{C}_4\text{H}_7\text{NO}_4$) (Li et al., 2013). Gu identified positive effects on the leaching rate in sulfuric acid after a thermal pretreatment of LiCoO_2 at 500°C (Gu and Nie, 2011). Joulié, Laucournet, and Billy (2014) applied selective precipitation from leaching solutions of hydrochloric, sulfuric, and nitric acids (Joulié, Laucournet, and Billy, 2014). Further leaching processes are discussed in detail by Li et al. (2009b, 2009c), Weng et al. (2013), and Kim et al. (2012).

3.5.2 Impurities and purification

In order to meet very high quality requirements, especially for usage in traction batteries, automotive original equipment manufacturers (OEMs) and their battery suppliers agree on very tight material specifications. If recycling processes are designed to gain raw materials in high quality, which can be reused to synthesize battery materials, these specifications have to be met. Thus, impurities stemming from the mixture of materials and various electrochemical reactions during the battery's life as well as impurities from the recycling process itself have to be reduced to a minimum to avoid a downcycling of these high tech materials. Impurities could be organic components such as organic carbonates of the electrolyte and their decomposition products, graphite, and carbon black, inorganics such as copper and aluminum from current collector foils, iron, fluorine compounds, and other additives from the electrolyte such as magnesium.

Krüger et al. propose that especially high aluminum contents in leaching solutions disturb material conditioning and therefore influence the secondary particle shape in an undesired way. The electrochemical cycling performance of recycled material from spent Li-ion batteries at 1°C is slightly decreased, caused by lower electrode kinetics (Krüger et al., 2014). Weng et al. find indications that the presence of magnesium up to $x = 0.01$ has no significant impact on the overall performance of $\text{Li}[(\text{Ni}_{0.33}\text{Co}_{0.33}\text{Mn}_{0.33})_{1-x}\text{Mg}_x]\text{O}_2$ and state that magnesium impurities up to a level as high as 360 mg/L in leachate remains tolerable (Weng et al., 2013).

Ferreira realizes a selective preleaching step with NaOH for the separation of Al. Therefore, Ferreira et al. dissolved

battery electrode compounds (NMC-coated aluminum foils) even without a mechanical separation step. NaOH was "found very selective for the separation of Al, thus leaving all Co and majority of Li in the solid phase" (Ferreira et al., 2009). While Al leaching without a preliminary mechanical separation will not be an industrial applicable process due to process costs, it could still be used to remove less concentrated Al impurities if the bought-in cations do not result in additional impurities. Gratz et al. have already included that process in an experimental set-up (Gratz et al., 2014).

In order to purify the leach from copper, Li and partners describe a process in which Cu is removed through replacement by iron powder at first and the iron is precipitated using the "goethite" method afterward. By this method, 99 wt% of Cu and Fe could be removed going along with only 2–3% loss of Ni, Co, and Mn (Li et al., 2009b, 2009c). Gao and Gratz use fractional precipitation to remove Cu and Fe from leaching solutions as well (Gratz et al., 2014; Gao and Li, 2012).

3.5.3 Electrochemical processes

Ra and Han (2006) applied an electrochemical process called *Etoile-Rebatt* technology to regain LiCoO_2 . Separated LiCoO_2 was leached in a solution of LiOH and KOH, codeposited on a platinum working electrode, and precipitated again to regain high quality raw materials for battery active material synthesis (Ra and Han, 2006).

Mixtures of cobalt and lithium ions can also be separated using a bipolar membrane electrodialysis with ethylenediaminetetraacetic acid (EDTA) chelation. The selectivity, found by Iizuka in 2013, for lithium and cobalt is 99% (Iizuka et al., 2013).

3.5.4 Solvent extraction

Cobalt can also be regained by well-established solvent extraction technologies using bis(2,4,4-trimethylpentyl)phosphinic acid (Cyanex 272) as extractant for instance (Jha et al., 2013; Granata et al., 2012b). Manganese can be preliminarily removed using bis-2-ethylhexylphosphoric acid (D2EHPA) (Granata et al., 2012a).

3.5.5 Resynthesis and further processes

Further research about hydrometallurgical processes and processes to obtain high quality raw materials for battery active material synthesis from metal solutions can be found in the studies by Swain et al. (2007), Sun and Qiu (2012), Gao and Li (2012), Li et al. (2008), Ma et al. (2013), Zhu et al. (2012), and Zou et al. (2013).

3.6 Pyrometallurgical processes

Pyrometallurgical processes to recycle batteries were developed to regain metals from various battery chemistries. Copper, cobalt, and nickel can be regained after further processing of smelter output via leaching and solvent extraction. On the one hand, these processes are reliable for mixtures of different batteries and can be combined with the virgin material process route. Xstrata adds cobalt- and nickel-containing battery scrap as secondary feedstock material in the extractive cobalt, nickel, and copper process routes (Georgi-Maschler *et al.*, 2012). On the other hand, high temperature smelting results in high energy consumption and the previously addressed need for sophisticated gas scrubbing. Furthermore, pyrometallurgical recovery of lithium and aluminum is currently not possible (Li *et al.*, 2013). The latter caused Maschler to propose a combination of pyrometallurgical and hydrometallurgical processes in order to regain lithium in addition to cobalt-based alloys from spent LIBs. The Co alloy could serve as a master alloy for these commercial cobalt-based superalloys (Georgi-Maschler *et al.*, 2012). Elwert *et al.* examined the phase composition of high lithium slags and found that hydrometallurgical lithium recovery could be reasonable for certain slags from pyrometallurgical LIB recycling processes (Elwert *et al.*, 2012).

Another application of heat treatment is the direct resynthesis of LiCoO_2 of the recycled Co_3O_4 and Li_2CO_3 . Lu mixed these components in a ball-mill followed by a heating step at 750°C for 15 h (Lu *et al.*, 2013).

4 DANGERS IN BATTERY RECYCLING

The danger characteristics in recycling of LIBs could be classified roughly into three sections: electrical dangers, fire and explosion dangers, and chemical dangers. They cannot be observed separately but interact and influence each other. Electrical dangers arise from high voltage and current possibly causing serious injuries. Short circuits of whole battery systems or only of a battery cell are extremely dangerous because the energy content is discharged in a very short time period. This could be caused in the event of improper handling of the units during discharging, disassembly, or storage. Deep discharging of LIBs can lead to polarity reversal accompanied by gas formation and inner pressure of the cell rising (Mähliß, 2012). If a short circuit occurs during discharging, disassembly, or storage, the generated heat delivers the energy to initiate an exothermal chemical reaction inside the battery cell, which in turn can trigger further reactions. This process is called *thermal runaway* (Wen, Yu, and Chen, 2012b). Resultantly, gaseous reaction products are formed. Measurements of

the gaseous products of venting LIBs during heat tests showed the evolution of hydrogen, carbon monoxide, carbon dioxide, methane, ethane, ethene, ethyl fluoride, and propene (Spotnitz and Franklin, 2002; Mikolajczak *et al.*, 2011; Golubkov *et al.*, 2014). In addition, toxic hydrogen fluoride (HF) is a possible product of reactions where fluorides of the binder or the electrolyte salt are involved due to short circuits of LIB cells (Mikolajczak *et al.*, 2011). Some of these gaseous products are also flammable and increase the danger of fire. Gas formation leads to rising of the inner pressure of battery cells and can cause explosions. During crushing of charged LIBs, short circuits occur due to the rotor cutting the battery cells, which brings the charged electrodes into electrical contact. To prevent short-circuiting during storage, disassembly, and crushing and thereby the formation of flammable and harmful gaseous products, it is advisable to deeply discharge the battery systems, modules, and cells. Process safety can be improved by processing these materials under an atmosphere with less moisture, avoiding hydrolysis of fluorinated conducting salts.

Today, electrolytes contain solvents such as DMC and EMC. These carbonates form explosive mixtures with aerial oxygen. For example, dimethyl carbonate has a lower explosion limit of 4.22 vol% (E.C.-E.C. Bureau, 2000). If LIBs do not undergo a thermal pretreatment, these components are released during crushing and afterward. To prevent fire or explosion, crushing under inert atmospheres is possible. In addition, the following process steps should be proceeded under inert conditions, until the solvents are removed. Ignition of the carbonates leads to high temperatures and decomposition of the commonly used fluorinated binder polyvinylidene fluoride, forming hydrogen fluoride (Ping *et al.*, 2014). Hydrogen fluoride is not only harmful but also can lead to increased corrosion on machines and security-relevant elements.

After several process steps, the coating materials of the electrodes exist as powder and contain high amounts of carbons. These powders have a high potential for dust explosion, which can be minimized by a tailored process design, but the risk of dust explosions should be evaluated for specific process products. The cathodic active materials, for example, lithium nickel cobalt manganese oxide, contain potential carcinogenic components such as nickel oxide (I.I.f.A.d.D.G. Unfallversicherung, 2014). Emission of these substances can be avoided by encasing machines and conveyance where these materials are processed. Regarding health and safety, workers have to be protected against emission during maintenance. There are the potential carcinogenic materials, as well as decomposition products of the electrolyte, such as fluorophosphates (Campion *et al.*, 2004).

5 INDUSTRIAL PROCESS CHAINS

Compared to smaller consumer batteries, the higher amount of valuable recyclables in larger battery systems used in EV and HEV increases the economical attractiveness for its recycling. Therefore, smaller- and industrial-scale recycling processes were developed over the past years. Georgi-Maschler (2009), Vezzini (2014), and Vadenbo (2009) give good overviews over these current industrial processes. As shown in Figure 3, all of those processes can be ascribed to a combination of the different unit operations: deactivation, mechanical treatment, hydrometallurgy, and pyrometallurgy. Figure 11 gives an overview of these combinations before the different processes will be illustrated in greater detail in the following sections.

5.1 Retrie Technologies Inc. (former Toxco Inc.)

Retrie Technologies Inc. (name change from Toxco in 2013) uses a combination of mechanical treatment and hydrometallurgical processing for the recovery of metals from spent LIBs. Toxco commercialized one of the first processes for the recycling of lithium batteries in 1992. The previous

development was motivated by the demand for a lithium battery recycling process in military systems (Miller and McLaughlin, 2001). In this former process, liquid nitrogen was used to deactivate the batteries and to reduce the reactivity of the metallic lithium during mechanical treatment (McLaughlin and Adams, 1999). This “is based on the principle that the kinetics of typical chemical reactions is halved when the temperature is decreased by 10 degrees. Therefore, at the temperature of liquid nitrogen (-273°C) the oxidation reaction rate is only $1/250000$ of the room temperature rate” (Miller and McLaughlin, 2001).

In 2009, the recycling of LIBs at Toxco’s plant in Ohio was granted by the US Department of Energy (Gaines *et al.*, 2011a). Today, the mechanical treatment (more precisely the comminution of the LIBs) is carried out with a crusher operating under a liquid solution. As a result of this, it is possible to reduce the reactivity of the processed batteries and to prevent volatile emissions. Larger battery packs (e.g., from hybrid and electric vehicles) first go through a manual disassembly process before the separated cells or smaller packs (i.e., laptop, power tool, and cell phone) are fed to the automated crusher (Retrie Technologies, Inc., 2014). The crushing is performed by a hammer mill while the brine

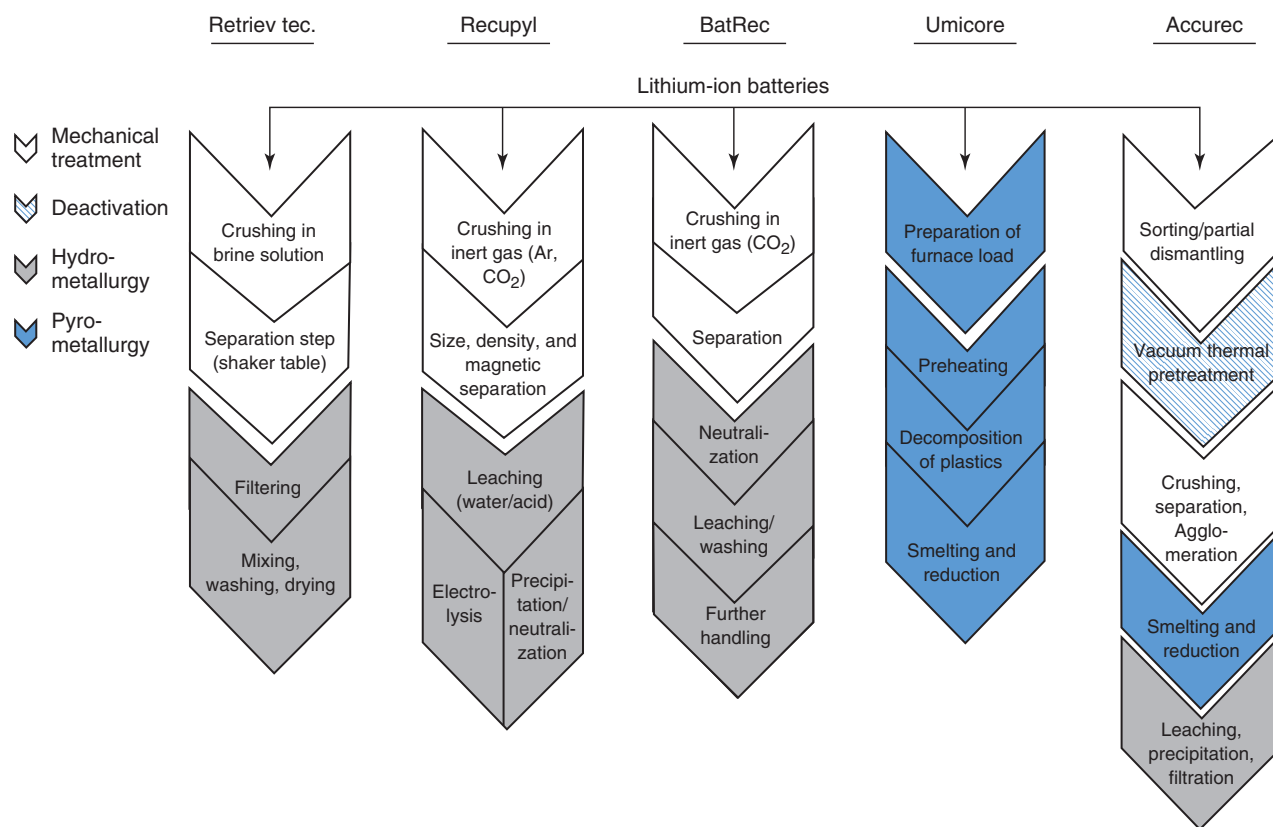


Figure 11. Overview of current industrial recycling processes for LIBs.

solution, used to accomplish a safer comminution, is recirculated from a downstream process step. Through this first mechanical process step, a so-called Li-ion fluff containing plastics and steel can be recovered. Depending on the amount of steel in this fluff, it could be disposed directly, or the steel could be recovered in a subsequent process. The next step, a mechanical separation process, is realized by a shaker table as seen in Figure 12. By injecting water and the movement of the table, two fractions can be recovered: a mixture of copper, cobalt, and aluminum and a slurry containing lithium, dispersed cobalt, and carbon. This remnant slurry is first put into a mix tank for further processing such as heating and reacting with added substances before it is transferred to a storage tank. Subsequently, the contained cobalt and carbon can be filtered out by means of a filter press. The emerging filter cake can be sold directly, or the cobalt can be recovered with higher purity through further metallurgical processing. The extracted Li-brine can either be recirculated to serve as an input for the wet crushing step or it can be transferred to a mixing tank where sodium carbonate (Na_2CO_3) is added. Adding Na_2CO_3 leads to the formation of insoluble lithium carbonate (Li_2CO_3) precipitating out of the solution. Afterward it can be filtered, washed, and dried (Vadenbo, 2009).

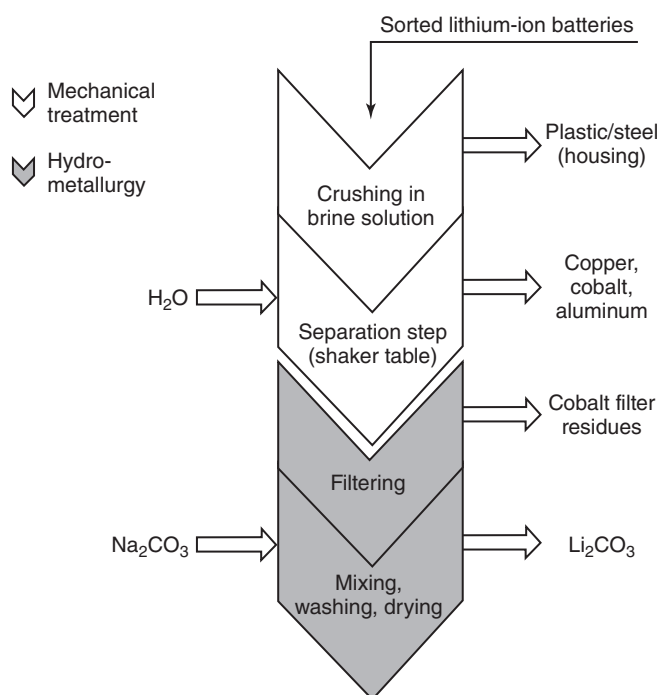


Figure 12. Flowchart of Toxco's (Retriev Technologies Inc.) hydrometallurgical recycling process for LIBs.

5.2 Recupyl

Recupyl SA, a French recycling company, was founded in 1993 with emphasis on the recycling of batteries. The underlying mechanical and hydrometallurgical process, shown in Figure 13, for the recycling of LIBs was developed during the VALIBAT project within the "European Commission's Fifth Framework Programme" (FP5) between 2000 and 2003 with Recupyl SA as coordinator (Vadenbo, 2009). According to Vadenbo, the process shown in the patent application 2007/0196725 A1 (Tedjar and Foudraz, 2007), which is designed for various lithium-based batteries, will be outlined concerning the recycling route for LIBs.

In a first step, the LIBs are crushed in an enclosure with defined and controlled atmosphere and pressure. The generated atmosphere consists of argon, carbon dioxide, or a mixture of these two gases (10–35 vol% argon) replacing the measured content of oxygen to preferred concentrations close to 5000 ppm. The comminution process is separated into two steps: the first one is realized in a slow shearing machine (preferably less than 11 rpm), whereas the second step is realized in a rotor system with a moderate speed (preferably less than 90 rpm). The comminuted material is then sent to a combined separation process consisting

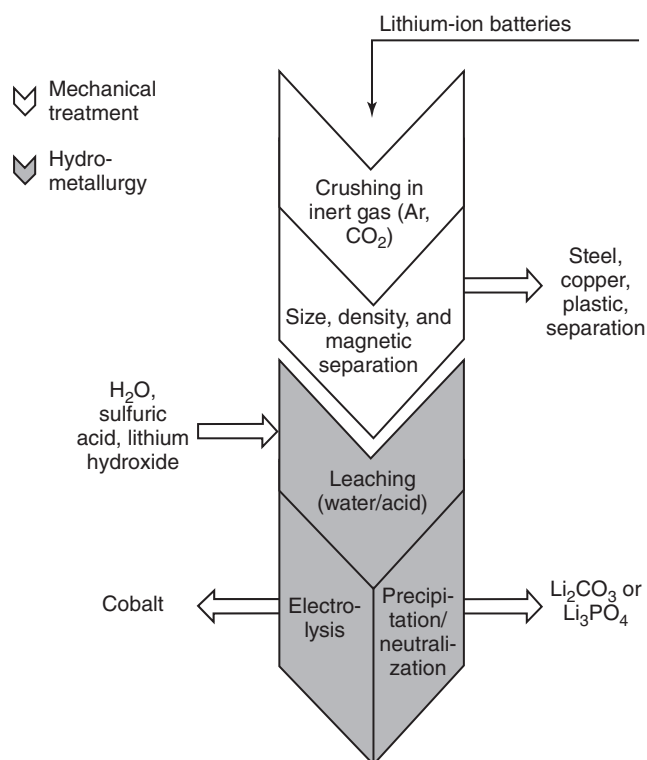


Figure 13. Flowchart of Recupyl's hydrometallurgical recycling process for LIBs.

of a vibrating screen, a magnetic separator, and a densimetric table that leads to four different fractions. The undersized fraction falling through the sieve mesh is rich in metal oxides and carbon; the magnetic fraction consists of stainless steel from the cell casing; the dense, nonmagnetic fraction consists of nonferrous materials, whereas the low density, nonmagnetic fraction includes separator materials and plastics. Following this, only the first mentioned undersized fraction is further processed by screening to reduce the content of copper to under 0.3%. Subsequently, the obtained fine fraction with high cobalt content can be processed in hydrometallurgical process steps by initial suspending in heavily stirred water. Here, the high turbulence leads to the release of metal and lithium oxides. By adding lithium hydroxide, the pH value increases and gaseous H_2 is generated by hydrolysis. At the end of the reaction and after a filtration step, a solution containing lithium salts and a suspension of metal oxides and carbon are obtained (Tedjar and Foudraz, 2007). By adding CO_2 , the lithium salt Li_2CO_3 is precipitated. Alternatively, a neutralization with phosphoric acid (H_3PO_4) leads to the formation of Li_3PO_4 . The mixture containing metal oxides and carbon can be processed by an additional acid leaching/filtration step with the result that cobalt can be obtained by means of electrolysis (Vezzini, 2014).

5.3 Batrec

The Batrec Industrie AG (Switzerland) belonging to the Veolia-Group is focused on the recycling of special waste with emphasis on batteries and mercury (Batrec Industrie AG, 2014). For the processing of spent LIBs, Batrec runs mainly a mechanical processing plant. Therefore, the batteries are conveyed by heatable or coolable screw-conveyors into the crusher where they are comminuted. To reduce the reactivity of the LIBs, this process step is conducted in a CO_2 atmosphere. Batrec denotes this as deactivation. The electrolyte evaporates during the crushing and is condensed in a subsequent step, though it is not considered for further use. The next process step is a mechanical separation that leads to four different fractions: two metals, one plastic, and one fine fraction containing the cobalt and lithium of the electrodes. After this step, the two metal fractions containing nonferrous metals and nickel can be provided to other recycling companies of the metal industry. The fine fraction is sold to cobalt- and nickel-producing companies such as Xstrata and Umicore, whereas the plastic fraction is partially used in pyrolysis processes for primary batteries (Georgi-Maschler *et al.*, 2012).

According to US patent application 2003/0180604 A1 (Zenger, Krebs, and Van Deutekom, 2003), a hydrometallurgical process is shown that connects to the mechanical

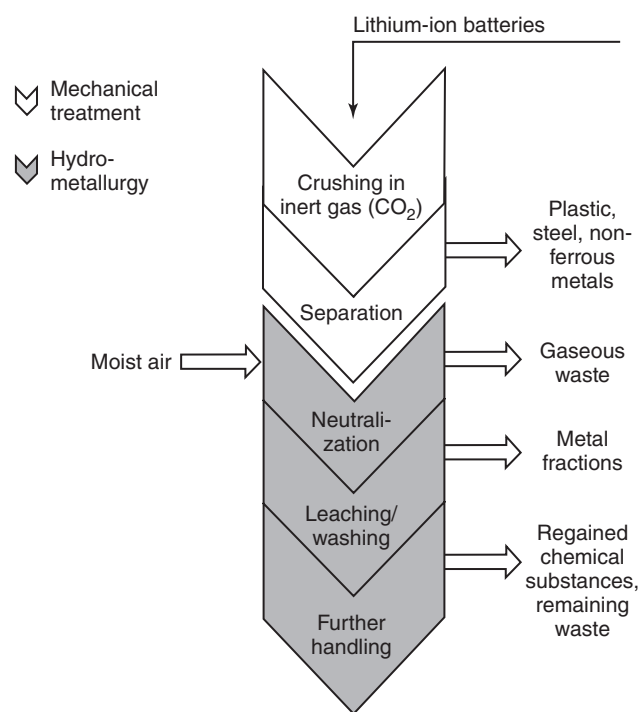


Figure 14. Flowchart of Batrec's hydrometallurgical recycling process for LIBs.

treatment described earlier. At first, a possible neutralization by adding moist air is implemented. The metal scrap then can be leached or washed using an acidified aqueous liquid (Figure 14), enabling metal fractions to be separated from the leaching liquor. For enhancement, a thermal treatment such as pyrolysis to remove residual plastics is mentioned. The leaching liquor can be processed in further steps to recover solute chemical substances (Zenger, Krebs, and Van Deutekom, 2003). Unfortunately, the patent application does not specify the substances or the types of residual waste leaving this process.

5.4 Inmetco

Inmetco (Pennsylvania), a subsidiary of Horsehead Holding Corp., recycles all secondary materials containing Fe, Ni, Co, Cr, and Mo. Thus, Inmetco owns a very flexible process allowing the acceptance of many different types of scraps (Cheret, 2007). They are specialized in the production of direct-reduced iron (DRI) out of remnants emerging in steel production. Since 2005, Inmetco runs a plant for recycling NiCd batteries, which was supplemented with the recycling of mercury-free zinc-carbon, NiMH, and also Li-ion batteries. However, zinc-carbon and NiMH batteries are fed into the DRI process where they are processed together with the steel remnants in a rotary hearth furnace, whereas the

NiCd batteries are processed in a separate process. From the resulting DRI, an NiCoCrFe alloy is produced in an electric arc furnace process that is fed with the spent LIBs donating the required cobalt. Consequently, this pyrometallurgical process just targets the recovery of cobalt and nickel from the LIBs. Most of the other contained materials just serve as reductants or are slagged so that this treatment is not a dedicated recycling process for LIBs (Georgi-Maschler, 2009).

5.5 Xstrata

Another possibility for the recycling of LIBs besides dedicated recycling processes is the feeding into the primary production of cobalt, nickel, or copper. Xstrata PLC, a Swiss mining company, adopts this approach by adding battery scrap as secondary feedstock material in the extractive cobalt, nickel, and copper process routes (Georgi-Maschler *et al.*, 2012). An advantage of this method is the reduction of energy consumption regarding the waste gas treatment. Because of the added sulfur-free battery scrap, the content of sulfur in the waste gases is reduced noticeably compared to the pure ore. There are also energy savings of about 75% because of the added secondary cobalt compared to the processing of the pure ore. In a first proceeding, entire LIBs were added to the process leading to increased moisture because of the contained electrolyte. This results in problems such as energy losses and soiling of plant sections. Since 2008, Xstrata implemented a pretreatment of the LIBs in a rotary kiln to dry out the electrolyte and to enable a higher throughput. Similar to Inmetco, the Xstrata process just targets the recovery of cobalt and nickel from the LIBs. Other contained materials just serve as reductants and are slagged or leave the process as waste gases (Georgi-Maschler, 2009).

5.6 Umicore

Umicore (Belgium), a global materials technology and recycling group, developed a new recycling concept called Val'Eas® in 2000–2003. This solution is dedicated for rechargeable NiMH and Li-ion batteries and is not derived from a process formerly used for other products. In comparison to other shown recycling processes, the spent batteries are directly fed into the pyrometallurgical process that is accomplished with a vertical shaft furnace. This means that there are no mechanical pretreatments such as shredding or separation steps except for partial dismantling, if needed, or the disassembly of big battery systems to the module level (Cheret, 2007). In a first step, a batch consisting of the batteries, slagformers, sand (SiO_2), coke, and limestone

(CaCO_3) is prepared. As shown in Figure 15, the furnace is divided into three sections: the preheating, the pyrolysis, and the smelting/reduction section (Vezzini, 2014). The prepared batch enters the furnace in the preheating section (lowest temperature) at its top. With temperatures of the charge below 300°C , the pressure inside of the batteries rises, and the evaporated electrolyte can leak. The reason for a desired slow temperature increase inside the batteries is the reduction in explosion risk. In the pyrolysis section, temperatures up to 700°C are high enough to melt down plastic components. Here, the high caloric value of the plastics increases the temperature of the gases needed for the preheating section. In the lowest section of the shaft furnace, further smelting and reduction processes take place. By injecting preheated air or even oxygen-enriched air, temperatures between 1200 and 1450°C are reached. At these temperatures, the remaining materials form two phases: a slag containing Al, Si, Ca, Li, and some Fe and an alloy containing mainly the metals Fe, Cu, Co, and Ni. Subsequently, the phases can be tapped in a granulated form (Cheret and Santén, 2008). Occurring waste gases leave the process at about 450°C . To avoid dangerous organic emissions such as furans or dioxins, the gases are treated in a thermal postcombustion (plasma) at temperatures from 3000 to 5000°C . The obtained metal alloys can be further processed by hydrometallurgical processes at the site in Olen (Belgium). Copper, iron, manganese, and zinc along

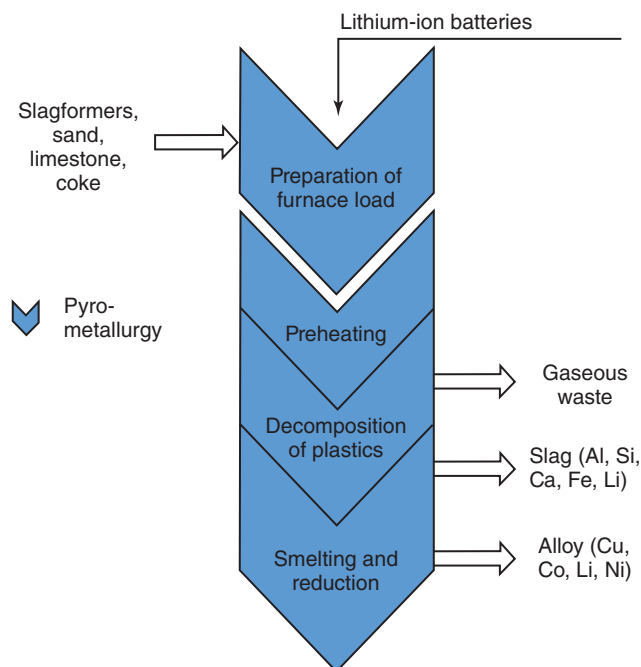


Figure 15. Flowchart of Umicore's pyrometallurgical recycling process for LIBs.

with nickel sulfate (NiSO_4) and cobalt chloride (CoCl_2) can be obtained using leaching, precipitation, filtration, and extraction methods (Georgi-Maschler, 2009). The formed slag can be reused in construction, concrete, or even in the steel industry (Cheret, 2007). According to Gaines *et al.*, Umicore claims a 93% recovery rate of LIBs, whereby a much smaller percentage actually comes out as usable high value material (Gaines *et al.*, 2011b).

5.7 Accurec

Accurec, a German battery recycling company founded in 1995, accepts all types of portable and industrial batteries for recycling. Because of increasing orders exceeding capacity, in 2014, a new site in Krefeld will be established. This new site is planned to recycle LIBs and also treat photovoltaic modules, whereas the old facility will keep up the recycling of NiCd and NiMH batteries (Accurec GmbH, 2014). According to Georgi-Maschler *et al.* (2012), the process for the recycling of LIBs developed by Accurec, the IME (University RWTH Aachen), and the project partner UVR-FIA GmbH combines mechanical pretreatment with hydro- and pyrometallurgical process steps including the possibility to recover lithium (Figure 16) (Georgi-Maschler *et al.*, 2012). In the first instance, a sorting of the LIBs is necessary to ensure that no other battery chemistries such as NiMH enter this process. After that, a disassembly step with the objective to obtain single cells leads to recovered housing materials such as plastics and also electronic components containing copper. The next process step is realized in a retort furnace at temperatures of up to 250°C and a vacuum of 650–700 mbar (vacuum thermal treatment). During this pyrolysis step, the battery cells are deactivated because the electrolyte evaporates and condenses in a downstream condenser. This should ensure a safer treatment in the following process steps. Unfortunately, the recaptured condensate contains various decomposition products that prevent direct reuse. Subsequently, the deactivated cells are crushed in a hammer mill. By means of a mechanical separation step (vibrating screen, magnetic separation, and air separation in a zigzag classifier), four fractions are obtained. A fine fraction containing the electrode materials is created in addition to three fractions consisting of iron-nickel, aluminum, and electrode foils. To prepare the fine fraction for the pyrometallurgical process, binder is added to agglomerate the particles to pellets. Because of the high amount of graphite (mass fraction of approx. 30%), a thermal pretreatment at 800°C reduces the content to about 15%, preventing a negative influence on the connecting pyrometallurgical process. Here, the pellets are treated in a carboreductive melting process resulting in a cobalt alloy, lithium-containing slag, and lithium-containing

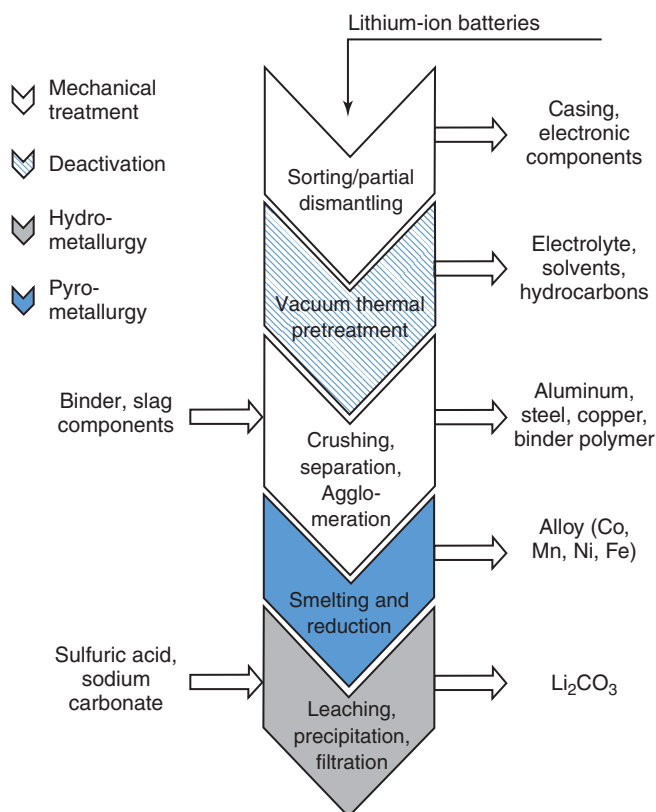


Figure 16. Flowchart of Accurec's recycling process for LIBs.

flue dust. The cobalt alloy can be sold as recycling product while the slag and the flue dust can be treated in additional hydrometallurgical steps. This is done by leaching, precipitation, and filtration operations with lithium carbonate (Li_2CO_3) as the final product with uses in batteries and the production of glass (Georgi-Maschler *et al.*, 2012; Georgi-Maschler, 2009).

5.8 Further processes for the recycling of LIBs

Another recycling process was developed by AEA Technology (UK). It consists of four main steps: mechanical shredding, electrolyte extraction, electrode dissolution, and cobalt reduction. The shredding of the cells is conducted in a dry and inert atmosphere to reduce the hydrolysis reaction of the electrolyte with water vapor from the air and to reduce possible implications of internal short circuits. Following this, the electrolyte can be extracted by immersing electrodes and separator fractions into a suitable solvent. The electrolyte can be recovered afterward in an evaporation step. To recover the electrode coating, the contained binder is also dissolved in a second dissolution step. This leads to detachment of the coating particles from the aluminum and copper foils so that

they can be recovered in a filtering step. In the last step, an electrochemical reduction of the suspended coating leads to the formation of solid cobalt oxide and lithium hydroxide (Lain, 2000).

OnTo Technology (USA) implemented the process invented by Steven Sloop (2003) in demonstration scale. The new approach of this process relies on the extraction of the electrolyte with a supercritical fluid (CO_2). Batteries with functional potential can be refurbished by refilling new electrolyte into the shells obtaining functional batteries. Understandably, the functionality of the refurbished cells needs to be tested. Cells with no functional potential or an insufficient performance level after the refurbishment can be treated by mechanical/physical crushing and separation steps to recover valuable materials (Vadenbo, 2009; Sloop, 2003).

6 RESEARCH PROJECTS LITHOREC AND LITHOREC II

The project LithoRec collaboration between industry partners AUDI, Chemetall, Electrocyling, Evonik Litarion, Lars Walch, H.C. Starck, I+ME ACTIA, Recyclx, Süd-Chemie, Volkswagen, and the universities TU Braunschweig and WWU Münster resulted in a combination of mechanical and hydrometallurgical process steps to recycle LIBs for traction applications. The project was funded by the German Federal Ministry for the Environment, Nature Conservation, and Nuclear Safety. The newly developed process route allows for the recovery of active materials, especially from $\text{LiNi}_{0.33}\text{Co}_{0.33}\text{Mn}_{0.33}\text{O}_2$ (NMC) electrodes.

At first, a disassembly system for LIB systems was designed (Herrmann *et al.*, 2014b), and the possibility to mechanize or automate several disassembly steps in order to separate the valuable battery cells or active cell materials was evaluated. In addition, a flexible gripper system with integrated functionalities such as voltage or resistance measurement was developed to show how the disassembly process can be supported by automation (Schmitt *et al.*, 2011b).

Furthermore, the evaluation of a mechanical and an alternative chemical recycling process for battery cell materials was carried out. The studies used basic process engineering principles to mechanically separate the current collector from the particle coating. These studies showed that by varying processing parameters the coatings were separated from the current collector with a yield of over 95% at a purity level that met the quality standards necessary for a direct recoating process (also applicable for production rejects) and further hydrometallurgical treatments (Hanisch, Haselrieder, and Kwade, 2011a, 2011b). In Figure 17, an



Figure 17. Intermediate and final recycling products of the combined mechanical and hydrometallurgical processes of LithoRec.

overview on the intermediate and final products gained from the combination of mechanical and hydrometallurgical treatment is given. In addition to electrochemical characterization, the adhesive strength and crystalline structure of recycled electrodes were analyzed and compared to those of reference electrodes made from commercial, nonrecycled materials. Laser diffraction spectroscopy, scanning electron microscopy, and atomic absorption spectroscopy analyses were conducted to determine particle size distribution, the structure of recycled electrodes, and the aluminum contamination of recovered active materials. As a quality check, the aging behavior of recycled and new-material electrodes was compared. The aging study showed that minor recycling impurities do not affect the electrochemical performance, whereas higher impurity levels significantly decrease electrochemical performance (Krüger *et al.*, 2014).

A strategic framework for the design of recycling networks for spent LIBs from EVs was developed. A mathematical optimization model shows the ideal theoretical evolution of recycling sites, capacities, and processes over time (Hoyer, Kieckhäfer, and Spengler, 2011). In addition, the impact of mandatory recycling rates on the recycling of LIBs from electric vehicles in Germany was determined. It was shown that a mandatory recycling rate of 50% results only in minor

advancements toward the desired recycling rates, but results in higher risk and lower financial attractiveness for investors (Hoyer, Kieckhäfer, and Spengler, 2013).

As a continuation to project LithoRec, LithoRec II proceeds to further investigate the aforementioned sustainable recycling process for LIBs. The main goal of the LithoRec process is to achieve a high recycling yield and to efficiently recover all battery materials at battery-grade quality. Thus, in addition to copper, cobalt, and nickel recycling, LithoRec also focuses on the recycling of lithium and electrolyte components.

LithoRec II spans the entire recycling process starting from the discharging and disassembly stages and following through with mechanical, chemical, and thermal processing (e.g., crushing, grinding, classification, sorting, and extraction) used to separate the battery into recyclable fractions. The project aims at a realization of the process chain in a pilot plant.

At the resulting pilot plant, the most valuable battery components, those which contain the battery active materials, are hydrometallurgically processed into preproducts of new battery materials. In addition, ecological and economical balances of the whole process are determined.

The companies of the LithoRec II consortium are Rockwood Lithium, Audi, Bosch Rexroth, ElectroCycling, H.C. Starck, Hosokawa Alpine, I+ME Actia, KUKA Roboter, Lion Engineering, Solvay Fluor, and Volkswagen. Research partners include the Institute for Particle Technology (iPAT), the Institute of Machine Tools and Production Technology (IWF), the Institute of High Voltage Technology and Electric Power Systems (elenia), the Institute for Chemical and Thermal Process Engineering (ICTV), the Institute of Automotive Management and Industrial Production (AIP) (all TU Braunschweig), as well as the MEET battery research center at Münster University.

7 CONCLUSION

According to a broad majority of experts, the recycling of LIBs is a crucial part of a sustainable strategy for all sorts of future energy storage applications. In the past, recycling of cobalt, nickel, and copper was the main focus of LIB recycling processes due to economic reasons. However, a future shortage of lithium is predicted within the next 100 years for scenarios without the application of recycling processes regaining 90% of the used lithium.

In order to recycle LIBs, a deactivation step should be chosen to lower the dangers resulting from the stored energy and chemical reaction potential, at first. Thermal pretreatment, discharging, and freezing can prevent electrochemical reactions and short circuits during a subsequent

crushing process. The following operations are divided into mechanical, pyrometallurgical, and hydrometallurgical treatments. Mechanical treatment implies crushing to open battery cells or modules followed by the liberation of the valuable materials, which is accompanied by classifying and sorting processes to separate copper foil, aluminum foil, separator, and the coating materials. During pyrometallurgical processes, several components of the battery cells or the battery cells themselves are melted. While the transition metals nickel, cobalt, and copper can be recycled from the cast, lithium and aluminum remain in the slag. Hydrometallurgical processes in battery recycling include leaching of the educts, extraction, crystallization, and precipitation. Hydrometallurgical treatment is used to recover pure metals, for example, lithium, from separated coating materials derived from mechanical processes or from slag derived from pyrometallurgical processes. The latter hydrometallurgical slag treatment is a difficult process chain.

Designing aforementioned processes, engineers have to be aware of great dangers in lithium-battery recycling resulting from the electrochemically stored energy, exothermic reactions, and occurring potentially dangerous reaction products.

Although a broad variety of processes developed by industry and by several researchers, such as Retrie Technologies, Batrec, Inmetco, Xstrata, Umicore, Accurec, Lion Engineering, and the LithoRec consortium, could be presented, serious efforts have to be made to recycle the necessary 90% of lithium in high purity economically and ecologically reasonable in the future.

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ENDNOTES

1. Assuming that a 30/70 (55%) V/R content for LiCoO_2 , graphite, steel, aluminum, and plastic can be used to make this Li-ion battery, an Emp value of 61 MJ/kg

was calculated. This amounts to a 37% reduction over the virgin value, again illustrating the energy benefit of using recycled materials in rechargeable batteries. If for these same materials only recycled stock is used, Emp becomes 45 MJ/kg, which amounts to a 54% reduction (Sullivan and Gaines, 2012).

2. The recycling processes promise significantly less energy and material demand than using primary raw materials. On the basis of the information reported by Dewulf *et al.* (2010), the recycling of NMC-active material needs only 15.7 MJ/kg NMC energy, while the production from primary materials calls for 25.1 MJ/kg NMC energy. The difference is significant; however, more detailed investigations are required. The share of energy consumption of raw materials and component production in entire energy demand of battery pack production is 21.6%. If secondary materials for production of NMC have been used, this would be decreased to 15.9%. This change of 5.5% means a decrease in the entire energy demand by 5.7% (Simon and Weil, 2013).

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